

Predicting Cardiovascular Disease Using Questions

Cardiovascular disease (CVD) includes conditions such as heart attack, stroke, coronary artery disease, and heart failure. Risk prediction using questions focuses on age, lifestyle, medical history, symptoms, and family background. This method does NOT diagnose CVD but helps estimate risk and suggest when medical checkups and tests are needed.

Key Categories for Cardiovascular Disease Screening

Category	Purpose	Example Questions
Demographics	Age, sex, BMI	What is your age, height, and weight?
Lifestyle	Smoking, alcohol, activity	Do you smoke, drink alcohol, or exercise regularly?
Diet	Fat, salt, processed food	Do you eat fried, oily, or salty foods frequently?
Symptoms	Heart-related signs	Do you experience chest pain, shortness of breath, or palpitations?
Family History	Genetic risk	Has anyone in your family had a heart attack or stroke?
Comorbidities	Linked conditions	Do you have diabetes, hypertension, or high cholesterol?
Stress & Sleep	Impact on heart health	Do you have chronic stress or poor sleep quality?

Common Warning Signs of Cardiovascular Disease

Warning signs that may suggest cardiovascular risk include:

- Chest pain, pressure, or tightness (especially with exertion)
- Shortness of breath during normal activity or at rest
- Pain in jaw, neck, back, or arms associated with exertion
- Palpitations (feeling like your heart is racing or skipping beats)
- Swelling in legs or ankles
- Sudden dizziness, weakness, or fainting
- Sudden difficulty speaking, facial drooping, or weakness on one side (possible stroke signs)

Risk Scoring Pattern

Assign scores to responses:

- ✓ No risk factor / healthy habit: 0
- ✓ Mild risk factor (e.g., slightly overweight, occasional junk food, mild stress): 1–2
- ✓ Strong risk factor (e.g., smoking, diabetes, hypertension, strong family history, frequent chest discomfort): 3–5

Risk Levels:

0–3 → Low risk

4–7 → Moderate risk

8+ → High risk — suggest full cardiac evaluation (ECG, lipid profile, BP check, stress test, etc.)

Include direct questions like:

- Have you ever been told you have a heart problem or abnormal ECG?
- Have you been diagnosed with high cholesterol, high blood pressure, or diabetes?

- Do you get chest tightness or breathlessness when climbing stairs?

Important:

This questionnaire-based method is a screening and risk estimation tool only. It cannot diagnose cardiovascular disease. Any high-risk response or worrying symptom should lead to immediate medical consultation and proper cardiac testing by a healthcare professional.